Sai Kiran Goud Lingala

Dayton, OH | (937) 516-4650 | saikiran27.gl@gmail.com
saikiran4099.github.io | linkedin.com/in/saikiranlingala27 | github.com/SaiKiran4099

SUMMARY

M.S. Mechanical Engineer specializing in computational fluid dynamics and aerodynamics with hands-on experience across ANSYS Fluent, OpenFOAM, and Simcenter STAR-CCM+. Skilled in RANS turbulence modeling, conjugate heat transfer, compressible flow validation, GCI-based uncertainty quantification, and Python-driven simulation automation. Proven ability to translate CFD results into actionable design decisions across propulsion, aerodynamic, and thermal systems.

SKILLS

CFD Tools: ANSYS Fluent, OpenFOAM, Simcenter STAR-CCM+, ParaView, ANSYS Meshing, ANSYS SpaceClaim
Analysis Methods: RANS (k-omega SST, k-epsilon), MRF and sliding-mesh rotating machinery, conjugate heat transfer (CHT), compressible and incompressible flow, GCI/Richardson extrapolation, y+ mesh control, turbulence closure selection, uncertainty quantification
Programming: Python (NumPy, Pandas, PyTorch, Matplotlib, scripting/automation), MATLAB, Fluent journal file automation
Design Tools: SolidWorks, CATIA V5/V6, geometry cleanup and model defeaturing for CFD import

EXPERIENCE

CFD and Aerodynamics Engineer Jan 2022 – Jul 2023
Cyient Hyderabad, India
- Performed engineering analysis on complex thermal-fluid systems using ANSYS Fluent, developing CFD models and applying GCI/Richardson extrapolation to validate results within 0.8% deviation.
- Conducted root cause analysis of flow separation and performance issues through systematic post-processing, evaluated engineering data, and recommended practical geometry solutions improving performance by 7% CL and 11% L/D.
- Prepared professional engineering reports and technical documentation, translating simulation results into clear recommendations for multidisciplinary engineering teams.
- Developed Python automation scripts and Fluent journal file templates, reducing new-case turnaround by approximately 35% across multi-case analysis campaigns.

Graduate Teaching Assistant, Heat Transfer and Controls Aug 2023 – Aug 2024
University of Dayton Dayton, OH
- Taught undergraduate and graduate heat transfer coursework covering conduction, convection, thermal boundary layers, and internal flow, reinforcing strong fundamentals in heat transfer and engineering mechanics.
- Mentored 20+ students in systematic engineering data analysis and technical reporting practices.

PROJECTS

Backward-Facing Step CFD Validation: RANS Turbulence Model Benchmarking *Simcenter STAR-CCM+*
- Built and meshed a backward-facing step case (Reh=5100) in Simcenter STAR-CCM+ matching a published experimental-DNS benchmark, verifying mesh quality before comparing Realizable k-epsilon against SST k-omega on an identical mesh.
- Quantified each model's deviation from the benchmark (k-epsilon 17-20% low, SST 11-19% high), publishing full methodology as a public report (github.com/SaiKiran4099/bfs-turbulence-study).

Rushton Turbine Stirred-Tank CFD Benchmark: MRF vs Sliding Mesh *Simcenter STAR-CCM+*
- Built a two-region, 774k-cell trimmed/prism Rushton stirred-tank model and computed impeller Power Number via steady MRF and transient sliding mesh, initially finding a 2.1x over-prediction versus the published correlation despite 1.7% agreement between formulations.
- Refuted rotor-stator unsteadiness and turbulence bias by direct test, traced the cause to a 25% oversized impeller sweep, predicted the corrected torque analytically, then confirmed it experimentally, landing within 3% of benchmark and 0.74% of each other (github.com/SaiKiran4099/RotorLab-CFD).

Turbine Blade Conjugate Heat Transfer and Cooling Analysis *ANSYS Fluent / SpaceClaim*
- Developed CHT models coupling fluid and solid thermal domains for a high-temperature blade with film cooling and thermal barrier coatings, analyzing heat transfer, fluid flow, and thermal performance.
- Extracted wall heat flux and temperature distribution data across cooling configurations, producing structured trade-off analysis and engineering recommendations.

2D Navier-Stokes Physics-Informed Neural Network Solver *Python / PyTorch*
- Implemented a mesh-free PINN in PyTorch using automatic differentiation to enforce mass and momentum conservation for steady-state Navier-Stokes equations.
- Validated solver velocity fields against analytical benchmarks for laminar cylinder flow (Re 10), quantifying residual error across flow regimes. Code: github.com/SaiKiran4099/NavierStokes-PINN-2D

EDUCATION

University of Dayton, Dayton, OH M.S., Mechanical Engineering, Aug 2023 – Dec 2025
Focus: Computational Fluid Dynamics, Heat Transfer, Compressible Flows, Propulsion

MLR Institute of Technology, Hyderabad, India B.Tech., Aeronautical Engineering, Aug 2018 – Jul 2022
Focus: Fluid Mechanics, Thermodynamics, Aerodynamics, Aerospace Structures